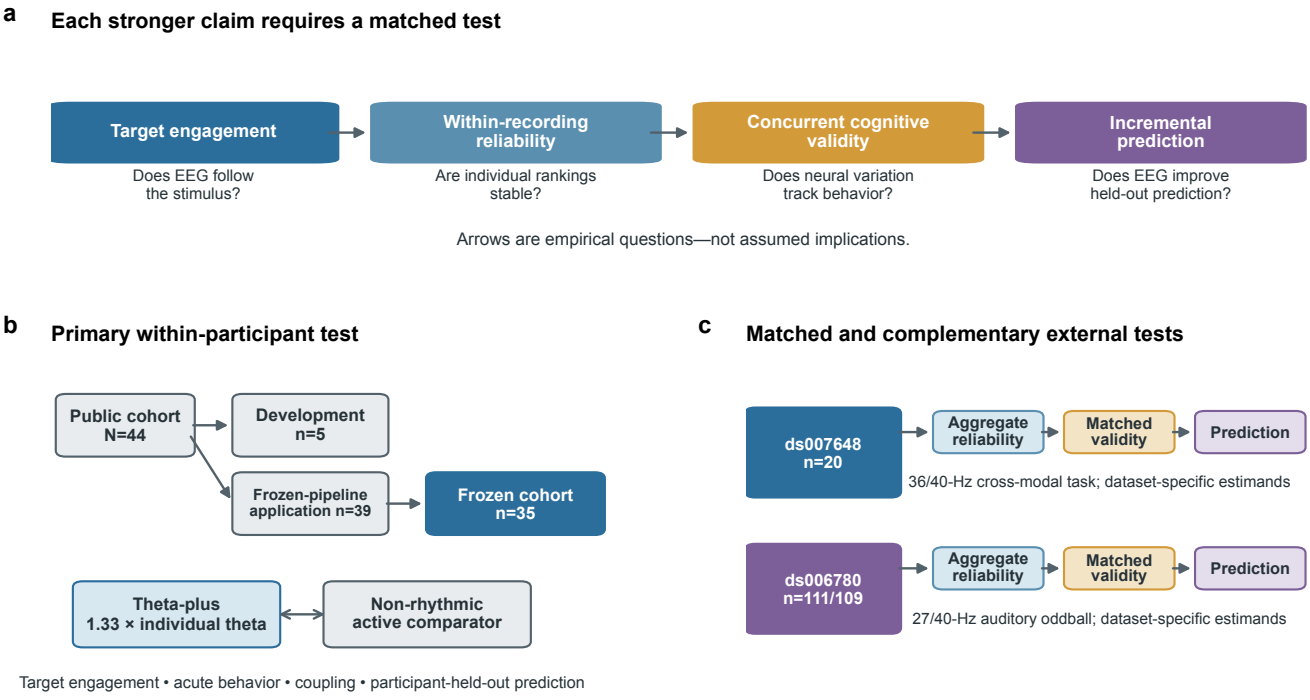

a
Each stronger claim requires a matched test
Target engagement
Does EEG follow the stimulus?
Within-recording reliability
Are individual rankings stable?
Concurrent cognitive validity
Does neural variation track behavior?
Incremental prediction
Does EEG improve held-out prediction?
Arrows are empirical questions—not assumed implications.
b
Primary within-participant test
Public cohort N=44
Development n=5
Frozen-pipeline application n=39
Frozen cohort n=35
Theta-plus 1.33 × individual theta
Non-rhythmic active comparator
Target engagement • acute behavior • coupling • participant-held-out prediction
c
Matched and complementary external tests
ds007648 n=20
Aggregate reliability
Matched validity
Prediction
36/40-Hz cross-modal task; dataset-specific estimands
ds006780 n=111/109
Aggregate reliability
Matched validity
Prediction
27/40-Hz auditory oddball; dataset-specific estimands